

# Adaptations in Athletic Performance after Ballistic Power versus Strength Training

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## ABSTRACT

CORMIE, P., M. R. MCGUIGAN, and R. U. NEWTON. Adaptations in Athletic Performance after Ballistic Power versus Strength Training. *Med. Sci. Sports Exerc.*, Vol. 42, No. 8, pp. 1582–1598, 2010. **Purpose:** To determine whether the magnitude of improvement in athletic performance and the mechanisms driving these adaptations differ in relatively weak individuals exposed to either ballistic power training or heavy strength training. **Methods:** Relatively weak men ( $n = 24$ ) who could perform the back squat with proficient technique were randomized into three groups: strength training ( $n = 8$ ; ST), power training ( $n = 8$ ; PT), or control ( $n = 8$ ). Training involved three sessions per week for 10 wk in which subjects performed back squats with 75%–90% of one-repetition maximum (1RM; ST) or maximal-effort jump squats with 0%–30% 1RM (PT). Jump and sprint performances were assessed as well as measures of the force–velocity relationship, jumping mechanics, muscle architecture, and neural drive. **Results:** Both experimental groups showed significant ( $P \leq 0.05$ ) improvements in jump and sprint performances after training with no significant between-group differences evident in either jump (peak power: ST =  $17.7\% \pm 9.3\%$ , PT =  $17.6\% \pm 4.5\%$ ) or sprint performance (40-m sprint: ST =  $2.2\% \pm 1.9\%$ , PT =  $3.6\% \pm 2.3\%$ ). ST also displayed a significant increase in maximal strength that was significantly greater than the PT group (squat 1RM: ST =  $31.2\% \pm 11.3\%$ , PT =  $4.5\% \pm 7.1\%$ ). The mechanisms driving these improvements included significant ( $P \leq 0.05$ ) changes in the force–velocity relationship, jump mechanics, muscle architecture, and neural activation that showed a degree of specificity to the different training stimuli. **Conclusions:** Improvements in athletic performance were similar in relatively weak individuals exposed to either ballistic power training or heavy strength training for 10 wk. These performance improvements were mediated through neuromuscular adaptations specific to the training stimulus. The ability of strength training to render similar short-term improvements in athletic performance as ballistic power training, coupled with the potential long-term benefits of improved maximal strength, makes strength training a more effective training modality for relatively weak individuals. **Key Words:** JUMP, SPRINT, SQUAT, NEUROMUSCULAR ADAPTATIONS

Improving athletic performance in already strong, well-trained athletes requires the development of sophisticated resistance training programs that contain a great deal of specificity and variability (15). In contrast, previously untrained individuals with low levels of strength display training-induced improvements in muscular function that are easily invoked and relatively nonspecific (38). In fact, improvements in a variety of athletic performance measures have been observed after periods of ballistic power training (10,14,18,19,25,39,40) as well as heavy strength training (20,34,36,38,39) in relatively weak, recreationally trained individuals. Although the training stim-

ulus varies considerably between these two modalities, short-term (i.e., 4–12 wk) exposure to either type of training has elicited neuromuscular adaptations necessary to improve strength, power, and speed in such individuals (10,13,14,18,38,39). Previous research has endeavored to examine the differential effects of these training modalities on performance of a range of athletic activities (18,39) as well as the effects of various loading conditions on the force–velocity relationship of single-joint movements (20). However, it remains unclear if improvements in athletic performance and the mechanisms driving adaptations in relatively weak individuals differ between ballistic power training and heavy strength training programs commonly used by strength and conditioning practitioners (i.e., typical load, repetition, and set combinations of complex, multi-joint, sports-specific movements). This information is essential to develop effective training programs for athletes with limited strength training background and to gain an enhanced understanding of the mechanisms responsible for performance improvements after training with sports-specific movements.

Ballistic power training is commonly used to target improvements in maximal power output and athletic performance. This training modality involves exercises that

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require the athlete to exert as much force as possible in short periods of time (i.e., ballistic movements), with the goal of projecting the accelerated object into free space (e.g., jumping, throwing, kicking). Improvements in performance after ballistic power training are common throughout the literature and typically include increases in maximal power output (10,19,25,40), rate of force development (RFD) (14,40), movement velocity (10,25,40), jump height (10,18,25,39), and sprint performance (19,25,39). Although such improvements are frequently observed by coaches and are reported in the research, it is less clear as to which detailed mechanistic factors drive these enhanced performances. Concurrent increases in maximal velocity or RFD and neural activation have been reported after ballistic power training (14). Strong theoretical arguments support the assumption that changes to neural activation patterns (i.e., lowered motor unit recruitment thresholds, improved motor unit firing frequency, and possibly synchronization) and enhanced intermuscular coordination contribute to the observed performance increases (14,32). However, these findings were made using isometric, single-joint movements (i.e., ankle dorsiflexion, elbow flexion/extension, knee extension). Similar results have not been consistently observed during dynamic, multijoint, sports-specific movements (25,39). This may be because of the distinct lack of research focus on the neuromuscular factors driving performance improvements in such movements after training. Furthermore, the extent to which other potential neuromuscular changes (i.e., muscle architecture, technique changes) contribute to improvements after ballistic power training is not clearly understood.

Strength training with heavy loads (i.e.,  $\geq 80\%$  one-repetition maximum (1RM)) has been commonly used to enhance athletic performance by improving the ability of the muscle to exert maximal force at any given velocity (15,20). Increasing maximal strength through such training has been shown to have a significant impact on athletic performance (20,34,36,38,39). Specifically, improvements in power output (20,36,39), movement velocity (20), and jump height (34,36,38,39) have been observed after periods of heavy strength training in relatively weak individuals. These changes occur because the skeletal muscle mechanics that give rise to the force-velocity relationship dictate that maximum strength plays a role in power production and thus influences athletic performance. Increased neural drive (13,25) and enhanced intermuscular coordination (32) are believed to be the primary mechanisms driving initial adaptations in relatively weak, inexperienced individuals exposed to heavy strength training (32). In addition, increased myofibrillar cross-sectional area (CSA) of both Type I and Type II fibers (although more pronounced in Type II fibers) are also evident after similar training interventions (16,23). These neuromuscular adaptations result in a shift in the force-velocity relationship in which muscle force is greater for any given velocity of shortening, thus resulting in more powerful muscular contractions (20).

Previous research has compared the ability of ballistic power training and heavy strength training to elicit improvements in athletic performance (18,39). In a study involving previously strength-trained men, Wilson et al. (39) reported ballistic power training (jump squats at 30% maximal isometric force) to transfer to improvements in countermovement jump (CMJ) and static jump (SJ) height, isokinetic torque at  $5.2 \text{ rad}\cdot\text{s}^{-1}$ , peak power during a 6-s cycle test, and a trend toward improved 30-m sprint time ( $P < 0.10$ ). Although heavy weight training (squats at 6–10RM) resulted in improved CMJ and SJ height as well as peak power during a 6-s cycle test. The percentage change in jump height for both CMJ and SJ was reported to be significantly greater for the ballistic power training group (39). Similarly, Harris et al. (18) observed improvements in CMJ height and the Margaria-Kalamen power (stair climb) test after training with a high force focus (i.e., strength training), whereas training with a high power emphasis elicited improvements in CMJ height, standing long jump, and the Margaria-Kalamen power test in previously strength-trained men. However, no significant differences were observed in the magnitude of performance improvements between the two groups (18). These results indicate that both training modalities do transfer to dynamic athletic performance after short-term interventions (9–10 wk) in moderately trained men and that ballistic power training may elicit greater improvements in athletic performance during this short period. Although these studies offer important insight into the transfer of training effect, they do not reveal any information regarding the mechanistic factors driving performance improvements. Therefore, it is unknown if or how the neuromuscular adaptations of relatively weak individuals differ after short-term ballistic power training or heavy strength training modalities. Furthermore, no research exists comparing the specific changes to the force-velocity relationship after ballistic power training versus heavy strength training (i.e., previous research has only examined the impact of strength training with various loads (20)).

Both ballistic power training and the initial stages of heavy strength training have commonly been associated with significant improvements in athletic performance and increased neural activation. Although the stimulus for adaptation differs considerably between these training modalities, the similarity of the mechanisms driving adaptation may influence the short-term changes in athletic performance observed in relatively weak or untrained individuals. Little research exists investigating the differential effects of ballistic power training versus heavy strength training on the force-velocity relationship (especially of sport-specific movements) as well as the transfer of training to athletic performance. Furthermore, there is a paucity of such research incorporating the examination of possible mechanisms responsible for changes in athletic performance. Therefore, the purpose of this experiment was to determine whether the magnitude of improvements in athletic performance and the mechanisms

driving these adaptations differ in relatively weak individuals exposed to either ballistic power training or heavy strength training.

## METHODS

**Experimental design.** This study used a randomized, control design and was conducted during a period of 15 wk. Subjects were randomly assigned to one of three groups: strength training group ( $n = 8$ ), power training group ( $n = 8$ ), or control group ( $n = 8$ ). The strength training group completed 10 wk of heavy squat training, the power training group completed 10 wk of ballistic jump squat training, and the control group maintained their normal level of activity throughout the duration of the study. Training involved three sessions per week in which subjects performed back squats at 75%–90% 1RM (strength training) or maximal-effort jump squats with 0%–30% 1RM (power training). Subjects completed a 2-d testing battery before initiating training (baseline), after 5 wk of training (midtest; strength and power training groups only), and after the completion of 10 wk of training (posttest). Seven days after the posttest occasion, subjects from the strength training group were tested for a fourth time (posttest 2) to assess changes in the force–velocity relationship independent of increased 1RM (i.e., jump squats performed at percentages of baseline 1RM). Subjects were thoroughly familiarized to all testing procedures before the actual assessment. Testing involved evaluation of jump and sprint performance as well as measures of the force–velocity relationship, jumping mechanics, muscle architecture, and neural drive.

**Subjects.** Male subjects who could perform a back squat with proficient technique were recruited for this study. Twenty-four men fulfilled all the testing and training requirements of this investigation (age =  $23.9 \pm 4.8$  yr, height =  $180.0 \pm 6.4$  cm, mass =  $79.8 \pm 12.0$  kg). Subjects' characteristics throughout the duration of the study are outlined in Table 1. The participants were notified about the potential risks involved and gave their written informed consent. This study was approved by the university's human research ethics committee.

**Training programs.** Both strength and power training programs involved three sessions per week, each separated by at least 24 h of recovery. Subjects refrained from any additional lower body resistance training, plyometrics, or sprint training outside the experimental training throughout the course of the study. The strength training group followed a program involving the back squat exercise exclusively. Each of the sessions was initiated with a warm-up consisting of 10 repetitions with an unloaded barbell (20 kg), 6 repetitions with 50% of that session's working resistance, and 4 repetitions with 70% of that session's working resistance. Adequate recovery was permitted between each warm-up set (3 min). Session 1 of the week involved three sets of three repetitions at 90% 1RM. Session 2 involved three sets of six repetitions at 75% 1RM. Session 3 involved three sets of four repetitions at 80% 1RM. All sessions involved an inter-set rest period of 5 min. These intensities were based on baseline 1RM values for the first 5 wk of training and midtest 1RM values for the last 5 wk of training. The load was reduced by 5% 1RM for the remainder of the session if subjects were unable to complete the required number of repetitions. Subjects were encouraged to perform each squat as rapidly as possible during the concentric phase while maintaining correct technique. Previous literature has shown significant improvements in maximal strength of relatively untrained participants after training with similar programming parameters (29).

Each power training session was initiated via a warm-up consisting of two sets of six submaximal jump squats with 0% 1RM (i.e., CMJ with no external load, just the resistance applied by a carbon fiber pole (mass = 0.4 kg) held across the shoulders). During sessions 1 and 3 of each week, subjects performed seven sets of six maximal-effort jump squats separated by a 3-min recovery. Jump squats were performed at the load that maximized power output for each individual subject, as determined during the baseline testing session. Similar to previous research, a load consistent with the subject's body mass (BM; i.e., no external load or 0% 1RM) maximized power output for each of the participants in this study (8,11). The second training session of each week included an additional warm-up set consisting of five submaximal jump squats with 30% 1RM. Subjects

TABLE 1. Subject characteristics of the strength training, power training, and control groups throughout the 10 wk of training.

|                   | BM (kg)         | Body Fat (%)   | 1RM/BM             | 40-m Sprint (s) | Jump Height (m) |
|-------------------|-----------------|----------------|--------------------|-----------------|-----------------|
| Strength training |                 |                |                    |                 |                 |
| Baseline          | 82.2 $\pm$ 13.7 | 16.7 $\pm$ 4.6 | 1.28 $\pm$ 0.17    | 5.86 $\pm$ 0.18 | 0.39 $\pm$ 0.04 |
| Midtest           | 82.8 $\pm$ 13.8 | 16.0 $\pm$ 4.6 | 1.57 $\pm$ 0.14*,† | 5.84 $\pm$ 0.15 | 0.42 $\pm$ 0.03 |
| Posttest          | 83.3 $\pm$ 12.8 | 15.9 $\pm$ 4.4 | 1.64 $\pm$ 0.13*,‡ | 5.76 $\pm$ 0.16 | 0.44 $\pm$ 0.06 |
| Power training    |                 |                |                    |                 |                 |
| Baseline          | 79.9 $\pm$ 14.5 | 17.3 $\pm$ 3.8 | 1.32 $\pm$ 0.14    | 5.90 $\pm$ 0.27 | 0.38 $\pm$ 0.04 |
| Midtest           | 79.2 $\pm$ 14.2 | 17.5 $\pm$ 4.0 | 1.38 $\pm$ 0.16    | 5.79 $\pm$ 0.20 | 0.42 $\pm$ 0.07 |
| Posttest          | 79.1 $\pm$ 13.8 | 17.7 $\pm$ 3.6 | 1.39 $\pm$ 0.17    | 5.69 $\pm$ 0.22 | 0.44 $\pm$ 0.06 |
| Control           |                 |                |                    |                 |                 |
| Baseline          | 77.5 $\pm$ 8.1  | 14.6 $\pm$ 3.5 | 1.37 $\pm$ 0.13    | 5.79 $\pm$ 0.19 | 0.41 $\pm$ 0.04 |
| Posttest          | 78.4 $\pm$ 8.8  | 14.7 $\pm$ 3.8 | 1.35 $\pm$ 0.12    | 5.79 $\pm$ 0.12 | 0.40 $\pm$ 0.04 |

\* Significantly ( $P \leq 0.05$ ) different from baseline.

† Significant ( $P \leq 0.05$ ) difference between strength training and power training groups.

‡ Significant ( $P \leq 0.05$ ) difference between strength training group and both power training and control groups.

then performed five sets of five maximal-effort jump squats with 30% 1RM separated by a 3-min recovery. Intensity was modified for each session so that an audible beep could be heard by subjects during jumps that reached 95% of the maximal power output at that load from their previous training or testing session. Previous literature has shown significant performance improvements after jump squat training with similar programming parameters at both 0% 1RM (10) and 30% 1RM (25,39).

**Testing protocol.** Subjects had 3–5 d of recovery between their previous training session and the midtest testing session. A period of 7–10 d of rest was required between the final training session and the posttest testing session in an attempt to maximize the response to the training intervention while minimizing fatigue (i.e., on the basis of the fitness-fatigue paradigm [35]). Subjects completed two testing sessions separated by at least 3 d of recovery at the baseline, midtest, and posttest occasions. Testing session 1 was initiated with an examination of the vastus lateralis (VL) muscle architecture through ultrasonography. Maximal dynamic strength was then assessed using a back squat 1RM to a depth consistent with a knee angle of at least 90° of flexion. During a 30-min recovery, body composition was assessed using dual-energy x-ray absorptiometry (DEXA). Maximal isometric strength was then evaluated using an isometric squat test performed at a knee angle of 140° to allow for the determination of maximum force output at zero velocity (11,28). Adequate recovery was permitted (10 min) before examination of jump squat performance across a series of intensities: 0% (i.e., no external load or BM only), 20%, 40%, 60%, and 80% of squat 1RM. SJ (i.e., no countermovement) performance was also assessed at 0% 1RM. Subjects completed the jump conditions (jump squats and SJ) in a randomized order, which was consistent across the three testing occasions for each individual subject. Kinematic (both linear position transducer (LPT) and two-dimensional motion analysis), kinetic, and electromyographic data were obtained simultaneously throughout the testing session. After at least 3 d of recovery, subjects completed the second testing session involving a 40-m sprint test. The strength training group completed an additional testing session (posttest 2) 7 d after the completion of the posttest occasion. This single testing session was identical to testing session 1 of all other test occasions except the jump squats were performed at relative percentages of *baseline* 1RM (i.e., 20%, 40%, 60%, and 80% of baseline 1RM). This additional testing session was used to assess the absolute changes in the force-velocity relationship independent of increases in 1RM.

**Data acquisition and analysis procedures.** The back squat 1RM involved subjects completing a series of warm-up sets (four to six repetitions at 30% of estimated 1RM, three to four repetitions at 50% of estimated 1RM, two to three repetitions at 70% of estimated 1RM, and one to two repetitions at 90% of estimated 1RM) each separated by a 3-min recovery. A series of maximal lift attempts was then performed until a 1RM was obtained. No more than

five attempts were permitted with each attempt separated by a 5-min recovery. This protocol has been frequently used throughout the previous literature for the assessment of maximal dynamic strength (8,10,11,25). Only trials in which subjects reached a knee angle of <90° of flexion were considered successful. This depth was visually monitored during testing and was confirmed by two-dimensional motion analysis (strength training group: baseline =  $86.0^\circ \pm 5.4^\circ$ , posttest =  $84.7^\circ \pm 3.7^\circ$ ; power training group: baseline =  $83.4^\circ \pm 2.4^\circ$ , posttest =  $80.4^\circ \pm 4.3^\circ$ ; control group: baseline =  $83.1^\circ \pm 6.2^\circ$ , posttest =  $82.6^\circ \pm 5.4^\circ$ ; rater reliability:  $r = 0.95$ ).

The isometric squat test was performed with subjects standing on a force plate (9290AD; Kistler Instruments, Winterthur, Switzerland) in a back squat position pushing against an immovable rigid bar. The bar was positioned so that subjects had a knee angle of 140° of flexion to allow for the determination of maximal force output at zero velocity. Previous research has shown this knee angle to correspond with the highest isometric force output for the squat compared with a range of other knee angles (28). Subjects were instructed to perform a rapid, maximal effort to reach maximal force output as soon as possible and maintain that force for 3 s. The analog signal from the force plate was collected for every trial at 1000 Hz using a data acquisition system including an analog-to-digital card (cDAQ-9172; National Instruments, North Ryde, NSW, Australia). Custom programs developed using LabVIEW software (Version 8.2; National Instruments, North Ryde, NSW, Australia) were used for recording and analyzing the data. The signal was filtered using a fourth-order, low-pass Butterworth filter with a cutoff frequency of 50 Hz. From laboratory calibrations, the voltage output was converted into vertical ground reaction force. Peak force relative to BM was assessed as the maximal force output during the 3-s period divided by the individual's BM. Test-retest reliability for peak isometric force relative to BM was  $r = 0.98$ .

Performance of a jump squat involved subjects completing a maximal-effort CMJ while holding a rigid bar across their shoulders. Subjects held a 0.4-kg carbon fiber pole for the 0% 1RM jump squat, whereas for all other intensities, subjects held a 20-kg barbell loaded with the appropriate weight plates. Participants were instructed to keep constant downward pressure on the bar throughout the jump and were encouraged to move the resistance as fast as possible to achieve maximal power output with each trial (2). Performance of the SJ also involved subjects holding the 0.4-kg carbon fiber pole across their shoulders. Subjects were instructed to lower into a back squat position with a knee angle of approximately 90° of flexion and hold this position for 3 s. When instructed, subjects then jumped as rapidly as possible in an attempt to maximize power output while performing no previous countermovement. Trials with any observable countermovement, determined as any degree of unloading in the vertical ground reaction force-time curve immediately before the jump, were repeated. For all jump

conditions, the bar was not to leave the shoulders of the subject, with the trial being repeated if this requirement was not met. A minimum of two trials at each load were completed, with additional trials performed if both peak power and jump height were not within 5% of the previous jump squat. Adequate rest was enforced between all trials (3 min).

All jump squats were performed while the subject was standing on a force plate with an LPT (PT5A-150; Celesco Transducer Products, Chatsworth, CA) attached to the bar. The LPT was attached 10 cm to the left of the center of the bar to avoid any interference caused by movement of the head during the jump. The LPT was mounted above the subject, and the retraction tension of the LPT (equivalent to 8 N) was accounted for in all calculations. Analog signals from the force plate and LPT were collected for every trial at 1000 Hz and analyzed using custom programs developed using LabVIEW software. The signal from the LPT was filtered using a fourth-order, low-pass Butterworth digital filter with a cutoff frequency of 10 Hz, and the voltage output was converted into displacement using laboratory calibrations. The vertical velocity of the movement was determined using a first-order derivative of the displacement data. Power output was calculated as the product of the vertical velocity and vertical ground reaction force data. Acceleration of the movement was calculated using a second-order derivative of the displacement data and was smoothed using a fourth-order, low-pass Butterworth digital filter with a cutoff frequency of 10 Hz. This data collection and analysis methodology has been validated previously (7), and test-retest reliability for all jump variables examined was consistently  $r \geq 0.90$ . Results have been expressed relative to BM.

A series of performance variables was assessed during the jump squats. Peak force, velocity, power, displacement, and acceleration were determined as the respective maximal values achieved during the entire movement. Net impulse was assessed as the integral of vertical ground reaction force during the period of application in which force exceeded that required during stationary standing (i.e., above body weight). RFD was determined between the minimum and maximum force that occurred throughout the movement. Similarly, rate of power development was also determined between the minimum and maximum power that occurred throughout the movement. Average power output was calculated during the concentric phase of the movement (i.e., time between minimum displacement and takeoff). Instantaneous force and velocity outputs at the time at which peak power occurred were also examined and were termed force at peak power and velocity at peak power, respectively. These values across each of the loads examined were used to generate the force-velocity and force-power relationships for the jump squat. Velocity at takeoff was defined as the velocity of movement at the time at which the force output was first zero (i.e., when the toes first left the force plate). Furthermore, time to takeoff was determined as the time between the initiation of the

countermovement (i.e., start of the eccentric phase) and the point that force was zero (i.e., end of the concentric phase—takeoff).

In addition to these instantaneous performance variables, analyses of parameters throughout the jump movement were conducted. The power-time and force-time curves from each individual subject were selected from the beginning of the eccentric phase (i.e., initiation of countermovement assessed as the time at which a change in velocity first occurred) through to the end of the concentric phase (i.e., at takeoff when force and power reached zero). The velocity-time and displacement-time curves were selected from the beginning of the eccentric phase to peak displacement (i.e., zero velocity). Using a custom-developed LabVIEW program, the number of samples in each individual curve was then modified to equal 500 samples by changing the time delta (dt) between samples and by resampling the signal (dt = number of samples in the original signal/500). The sampling frequency of the normalized signals was calculated according to the following equation:

$$\text{normalized sampling frequency (Hz)} = \frac{1 \text{ second}}{\left[ \frac{\text{no. samples in original signal}}{\text{no. samples in normalized signal}} \right] \times \left[ \frac{\text{seconds}}{\text{per sample}} \right]}$$

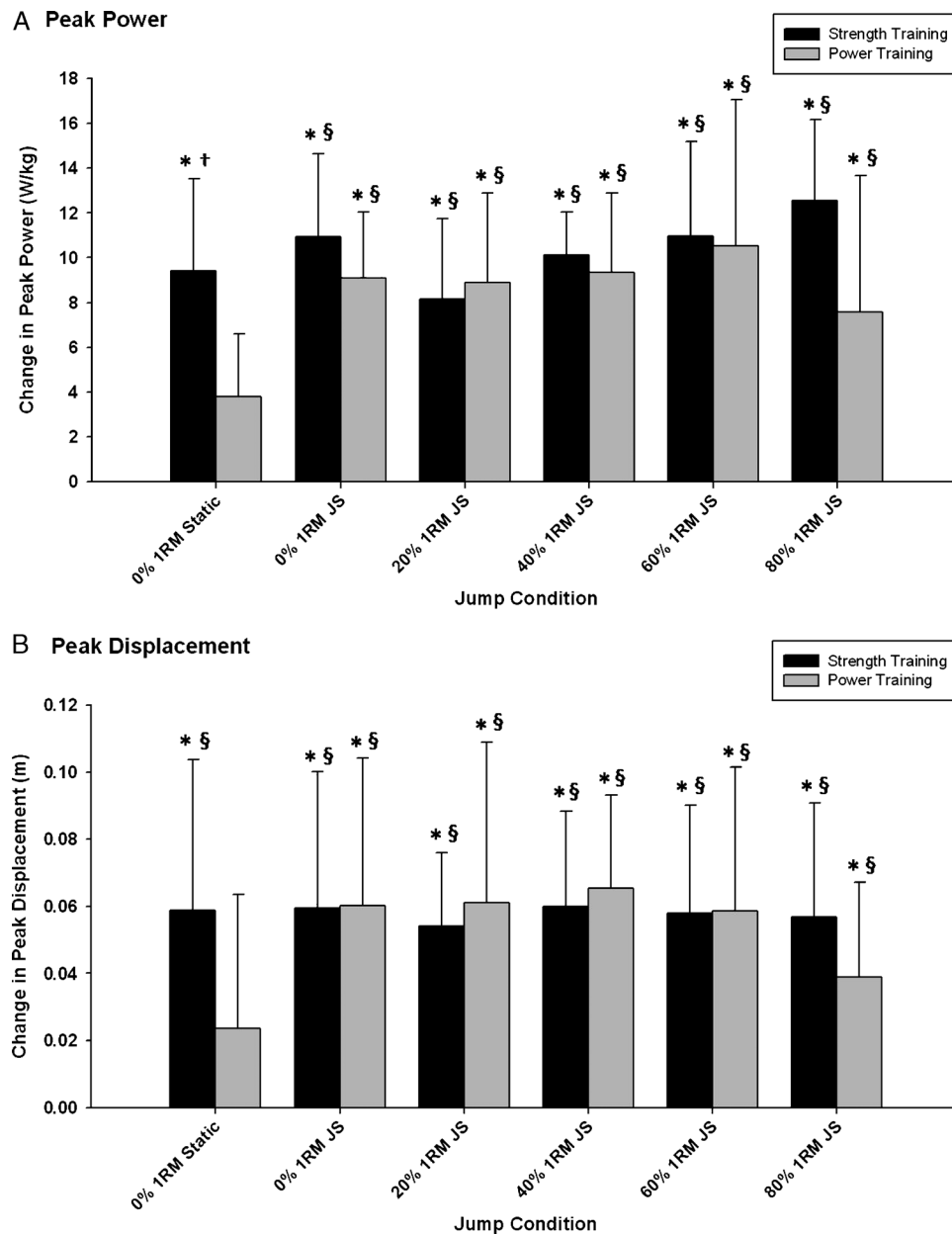
Consequently, the sampling frequencies of the modified signals were then equivalent to  $717 \pm 145$  Hz for the power-time and force-time curves and to  $501 \pm 74$  Hz for the velocity-time and displacement-time curves. This resampling allowed for each individual's power, force, velocity, and displacement curves to be expressed during equal periods of time (i.e., the 500 samples represented the relative time—from 0% to 100%—taken to complete the jump). In other words, the various data sets were normalized to time so that data could be pooled. Each sample of the normalized power-, force-, velocity-, and displacement-time curves was then averaged across subjects within the strength training, power training, or control groups, resulting in averaged curves with high resolution (sampling frequency of 501–717 Hz). This allowed for power, force, velocity, and displacement throughout the jump to be compared across the baseline, midtest, and posttest occasions as well as between groups. Intraclass test-retest reliabilities for power-, force-, velocity-, and displacement-time curves during the CMJ have consistently been  $r \geq 0.94$ ,  $r \geq 0.90$ ,  $r \geq 0.89$ , and  $r \geq 0.92$ , respectively, using this methodology (8,9).

Before assessment of sprint performance, subjects performed a warm-up consisting of 5 min of light jogging and three submaximal 20-m sprints. The sprint test was initiated from a standing start involving a staggered stance with the same front-back leg orientation used for every trial throughout the baseline, midtest, and posttest occasions. Subjects were instructed to commence the sprint at will and accelerate as quickly as possible throughout the 40 m. Three trials were performed, with each separated by a 3-min recovery. A series of six dual-beam timing gates (Speedlight;

Swift Sports, Lismore, NSW, Australia) recorded instantaneous time at 5, 10, 20, 30, and 40 m (timing was commenced when each subject passed through a dual-beam timing gate positioned at their front foot). Flying 5 m was calculated as the time between the 5- and 10-m gates, whereas flying 15 m was calculated as the time between the 5- and 20-m gates. Intraclass test-retest reliability for all sprint performance variables examined was consistently  $r \geq 0.90$ .

EMG of the VL, vastus medialis (VM), and biceps femoris (BF) was collected on the dominate leg during the isometric squat and all jumps. Disposable surface electrodes

(self-adhesive Ag/AgCl snap electrode, 2-cm interelectrode distance, 1-cm circular conductive area; product 272; Noraxon USA, Inc., Scottsdale, AZ) were attached to the skin over the belly of each measured muscle, distal to the motor point, and parallel to the direction of muscle fibers. A reference electrode was placed on the patella. The exact location of the electrodes relative to anatomical landmarks was marked on a sheet of tracing paper after the first testing session to ensure consistent placement in subsequent tests. Each site was shaved, gently abraded, and cleansed with alcohol before electrode placement to minimize skin impedance. Raw EMG



**FIGURE 1**—Change in peak power (*A*) and peak displacement (*B*) between baseline and posttesting sessions across a variety of jump conditions. The change reported for the strength training group is between baseline and the posttest 2 session in which the loaded jump squats were assessed with percentages of their baseline 1RM. JS, jump squat; Static, SJ. \*Significant ( $P \leq 0.05$ ) change from baseline. †Significant ( $P \leq 0.05$ ) difference between strength training group and both power training and control groups. §Significantly ( $P \leq 0.05$ ) different from the control group.

TABLE 2. Changes in performance variables during the 0% 1RM jump squat (i.e., BM only) from baseline ( $\Delta$ ).

| 0% 1RM Jump Squat                                                             | Strength Training Group |                      | Power Training Group       |                      |
|-------------------------------------------------------------------------------|-------------------------|----------------------|----------------------------|----------------------|
|                                                                               | $\Delta$ at Midtest     | $\Delta$ at Posttest | $\Delta$ at Midtest        | $\Delta$ at Posttest |
| Power                                                                         |                         |                      |                            |                      |
| Peak power ( $\text{W}\cdot\text{kg}^{-1}$ )                                  | $6.6 \pm 2.6^*$         | $10.9 \pm 3.7^*$     | $7.1 \pm 4.1^*$            | $9.1 \pm 2.3^*$      |
| Average power ( $\text{W}\cdot\text{kg}^{-1}$ )                               | $4.3 \pm 2.5^*$         | $7.6 \pm 2.5^*$      | $7.0 \pm 2.3^*,\dagger$    | $8.5 \pm 1.9^*$      |
| Rate of power development ( $\text{W}\cdot\text{kg}^{-1}\cdot\text{s}^{-1}$ ) | $117.9 \pm 105.0^*$     | $158.0 \pm 101.7^*$  | $169.3 \pm 79.4^*,\dagger$ | $236.7 \pm 129.7^*$  |
| Force                                                                         |                         |                      |                            |                      |
| Peak force ( $\text{N}\cdot\text{kg}^{-1}$ )                                  | $2.0 \pm 2.4$           | $3.0 \pm 2.6^*$      | $3.5 \pm 3.7$              | $4.7 \pm 4.2^*$      |
| RFD ( $\text{N}\cdot\text{kg}^{-1}\cdot\text{s}^{-1}$ )                       | $25.7 \pm 27.8$         | $39.2 \pm 28.6^*$    | $55.3 \pm 28.8^*,\dagger$  | $94.6 \pm 74.9^*$    |
| Force at peak power ( $\text{N}\cdot\text{kg}^{-1}$ )                         | $1.2 \pm 1.1^*$         | $1.6 \pm 1.0^*$      | $1.5 \pm 1.8$              | $1.4 \pm 1.9$        |
| Net impulse (N-s)                                                             | $32.2 \pm 26.9^*$       | $57.1 \pm 33.6^*$    | $48.8 \pm 26.5^*$          | $57.4 \pm 20.9^*$    |
| Velocity                                                                      |                         |                      |                            |                      |
| Peak velocity ( $\text{m}\cdot\text{s}^{-1}$ )                                | $0.22 \pm 0.19^*$       | $0.36 \pm 0.22^*$    | $0.34 \pm 0.34^*$          | $0.55 \pm 0.26^*$    |
| Velocity at peak power ( $\text{m}\cdot\text{s}^{-1}$ )                       | $0.16 \pm 0.21$         | $0.32 \pm 0.28^*$    | $0.15 \pm 0.30$            | $0.28 \pm 0.27$      |
| Velocity at takeoff ( $\text{m}\cdot\text{s}^{-1}$ )                          | $0.24 \pm 0.22^*$       | $0.37 \pm 0.23^*$    | $0.35 \pm 0.36^*$          | $0.60 \pm 0.30^*$    |
| Acceleration                                                                  |                         |                      |                            |                      |
| Peak acceleration ( $\text{m}\cdot\text{s}^{-1}\cdot\text{s}^{-1}$ )          | $4.1 \pm 3.9$           | $5.7 \pm 5.2^*$      | $6.4 \pm 2.0^*$            | $8.6 \pm 4.7^*$      |
| Displacement                                                                  |                         |                      |                            |                      |
| Peak displacement (m)                                                         | $0.03 \pm 0.01^*$       | $0.06 \pm 0.04^*$    | $0.04 \pm 0.05$            | $0.06 \pm 0.04^*$    |
| Time                                                                          |                         |                      |                            |                      |
| Time to takeoff (s)                                                           | $-0.18 \pm 0.16$        | $-0.20 \pm 0.21$     | $-0.27 \pm 0.12^*,\dagger$ | $-0.31 \pm 0.13^*$   |

\* Significant ( $P \leq 0.05$ ) change from baseline.† Significant ( $P \leq 0.05$ ) difference between strength training and power training groups.

signals were collected at 1000 Hz and amplified (gain = 1000, bandwidth frequency = 10–1000 Hz, input impedance < 5 k $\Omega$ ; Model 12D-16-OS; Neurodata Amplifier System, Grass Technologies, West Warwick, RI). The amplified myoelectric signal was collected simultaneously with the force plate and LPT data using a data acquisition system including an analog-to-digital card. Custom programs developed using LabVIEW software were used for recording and analyzing the data. The signal was full wave-rectified and filtered using a dual-pass, sixth-order, 10- to 250-Hz band-pass Butterworth filter as well as a notch filter at 50 Hz. A linear envelope was created using a low-pass, fourth-order Butterworth digital filter with a cutoff frequency of 6 Hz. Maximal voluntary contraction (MVC) for all muscles was determined by averaging the integrated EMG signal during a 1-s period

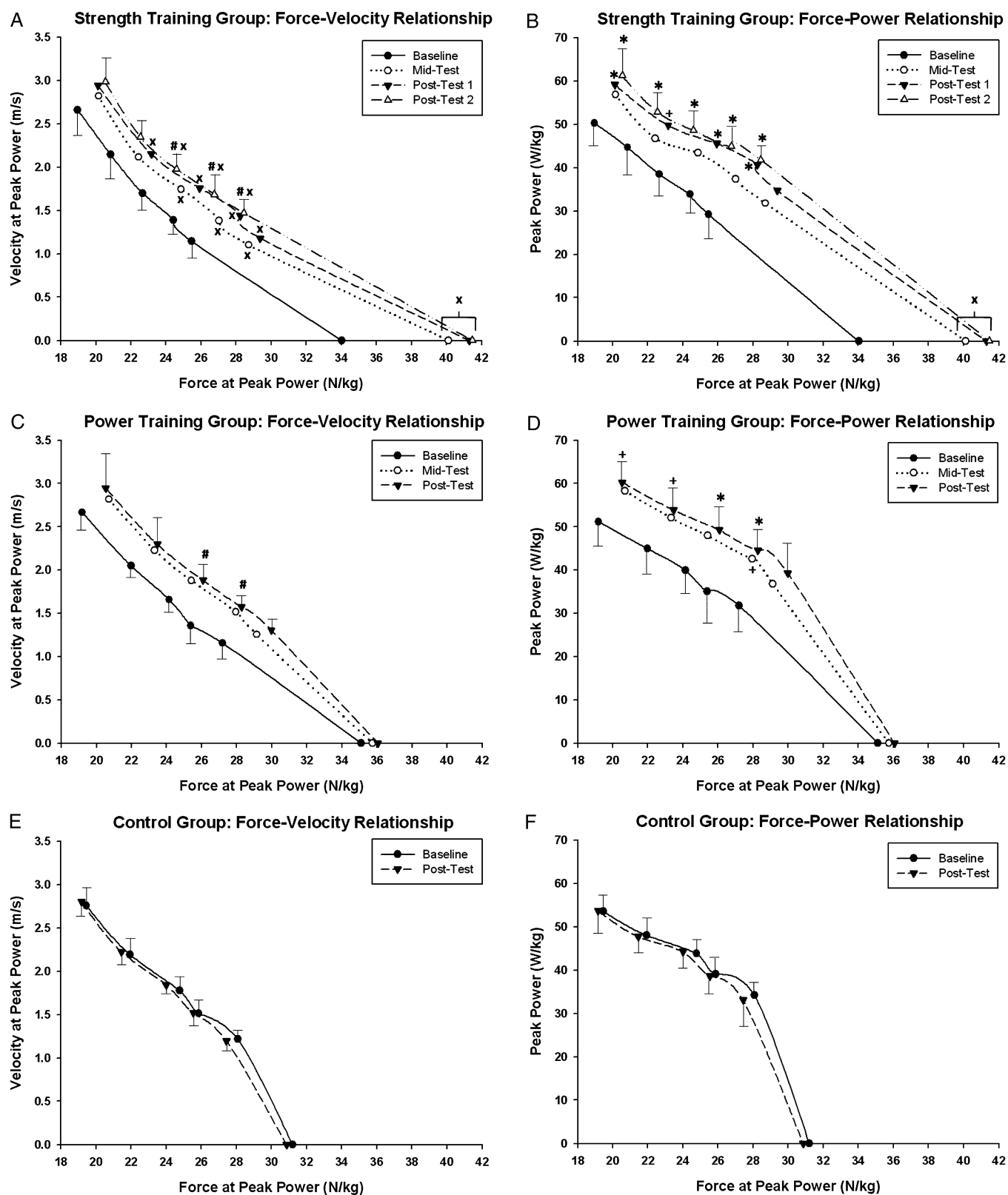
of sustained maximal force output after the initial peak in the force curve during the isometric squat (intraclass test-retest reliability consistently  $r \geq 0.91$ ). EMG activity during jumping was analyzed by averaging the integrated EMG signal from the beginning of the eccentric phase to takeoff. To standardize for the time taken to complete a jump squat, this value was then divided by the time to takeoff. The average integrated EMG (AvgIEMG) was then normalized by expressing it relative to the MVC. This is similar to methods previously used when comparing EMG between movements with different time components (27). The rate of rise in AvgIEMG (expressed as a percentage of MVC per second) was assessed during the 0% 1RM jump squat as the rate between the minimum and maximum AvgIEMG throughout the movement (i.e., from the initiation of the

TABLE 3. Comparison of 40-m sprint times among strength training, power training, and control groups throughout the 10 wk of training.

|                                        | Sprint Times (s) |                  |                            |                            |                            |                  |                            |
|----------------------------------------|------------------|------------------|----------------------------|----------------------------|----------------------------|------------------|----------------------------|
|                                        | 5 m              | 10 m             | 20 m                       | 30 m                       | 40 m                       | Flying 5 m       | Flying 15 m                |
| Strength training                      |                  |                  |                            |                            |                            |                  |                            |
| Baseline                               | $1.16 \pm 0.08$  | $1.92 \pm 0.10$  | $3.30 \pm 0.13$            | $4.59 \pm 0.16$            | $5.88 \pm 0.18$            | $0.76 \pm 0.03$  | $2.15 \pm 0.07$            |
| Midtest                                | $1.18 \pm 0.08$  | $1.96 \pm 0.10$  | $3.30 \pm 0.12$            | $4.57 \pm 0.14$            | $5.82 \pm 0.17$            | $0.79 \pm 0.03$  | $2.12 \pm 0.06$            |
| Posttest                               | $1.15 \pm 0.06$  | $1.92 \pm 0.08$  | $3.27 \pm 0.13$            | $4.53 \pm 0.15$            | $5.76 \pm 0.17$            | $0.78 \pm 0.03$  | $2.13 \pm 0.08$            |
| $\Delta$ between baseline and posttest | $-0.01 \pm 0.04$ | $0.01 \pm 0.04$  | $-0.03 \pm 0.05$           | $-0.06 \pm 0.07$           | $-0.12 \pm 0.06^*$         | $0.02 \pm 0.02$  | $-0.02 \pm 0.03$           |
| ES                                     | 0.14             | 0.08             | 0.25                       | 0.43                       | 0.67                       | 0.62             | 0.29                       |
| Power training                         |                  |                  |                            |                            |                            |                  |                            |
| Baseline                               | $1.17 \pm 0.12$  | $1.95 \pm 0.16$  | $3.32 \pm 0.19$            | $4.60 \pm 0.23$            | $5.90 \pm 0.27$            | $0.78 \pm 0.04$  | $2.14 \pm 0.08$            |
| Midtest                                | $1.13 \pm 0.05$  | $1.90 \pm 0.08$  | $3.24 \pm 0.11$            | $4.51 \pm 0.16$            | $5.79 \pm 0.20$            | $0.77 \pm 0.05$  | $2.10 \pm 0.09$            |
| Posttest                               | $1.09 \pm 0.05$  | $1.86 \pm 0.07$  | $3.18 \pm 0.12$            | $4.44 \pm 0.17$            | $5.69 \pm 0.22$            | $0.77 \pm 0.02$  | $2.09 \pm 0.08$            |
| $\Delta$ between baseline and posttest | $-0.08 \pm 0.09$ | $-0.09 \pm 0.10$ | $-0.13 \pm 0.09^*,\dagger$ | $-0.16 \pm 0.10^*,\dagger$ | $-0.22 \pm 0.14^*,\dagger$ | $-0.01 \pm 0.03$ | $-0.06 \pm 0.03^*,\dagger$ |
| ES                                     | 0.79             | 0.69             | 0.78                       | 0.76                       | 0.82                       | 0.25             | 0.66                       |
| Control                                |                  |                  |                            |                            |                            |                  |                            |
| Baseline                               | $1.13 \pm 0.04$  | $1.88 \pm 0.07$  | $3.25 \pm 0.11$            | $4.52 \pm 0.14$            | $5.79 \pm 0.19$            | $0.75 \pm 0.03$  | $2.12 \pm 0.08$            |
| Posttest                               | $1.15 \pm 0.04$  | $1.91 \pm 0.05$  | $3.27 \pm 0.07$            | $4.53 \pm 0.09$            | $5.79 \pm 0.12$            | $0.76 \pm 0.02$  | $2.12 \pm 0.05$            |
| $\Delta$ between baseline and posttest | $0.02 \pm 0.05$  | $0.03 \pm 0.06$  | $0.02 \pm 0.08$            | $0.01 \pm 0.08$            | $0.00 \pm 0.10$            | $0.01 \pm 0.02$  | $0.00 \pm 0.04$            |
| ES                                     | 0.41             | 0.46             | 0.24                       | 0.10                       | 0.01                       | 0.40             | 0.07                       |

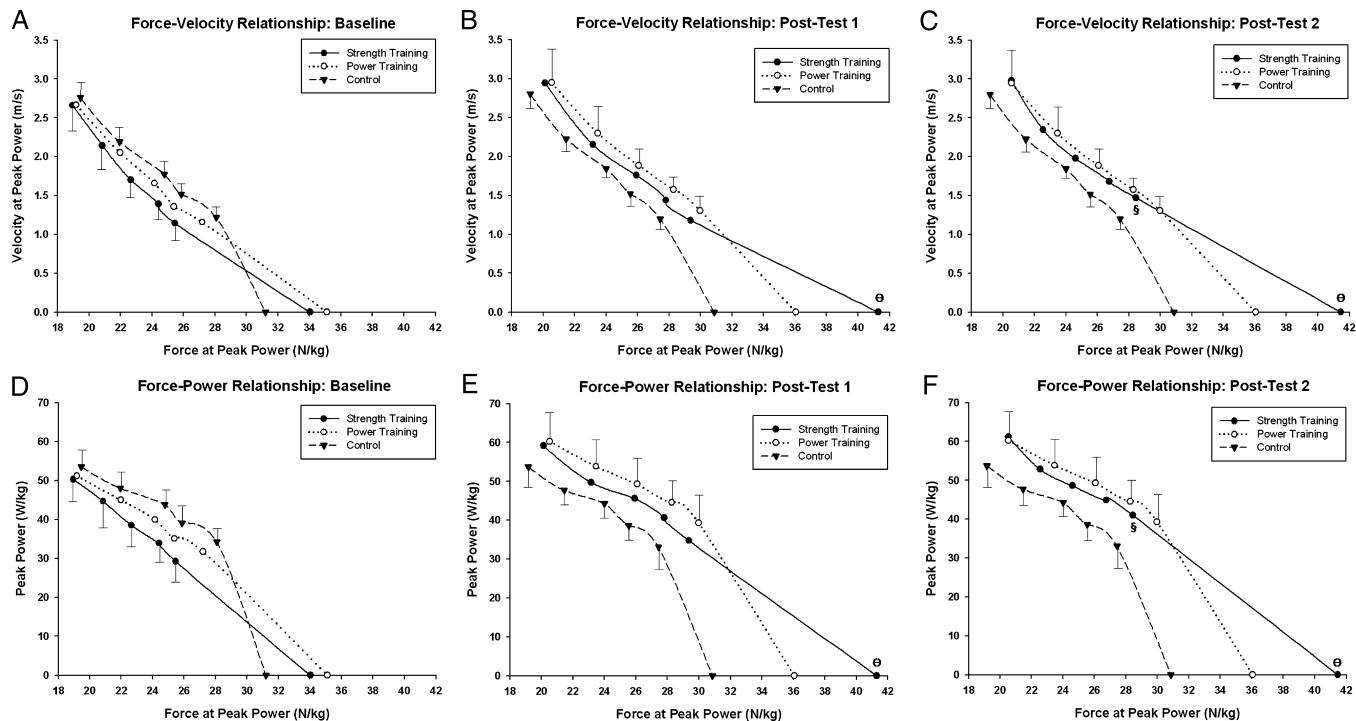
Flying 5-m time is calculated as time between 5 and 10 m. Flying 15-m time is calculated as time between 5 and 20 m.

The change in sprint times between baseline and posttests ( $\Delta$  baseline and posttest) as well as ES of the change is also displayed.\* Significant ( $P \leq 0.05$ ) change from baseline to posttest.† Significantly ( $P \leq 0.05$ ) different from the control group.



**FIGURE 2**—Training-induced changes to the force-velocity and force-power relationships for the jump squat in the strength training (*A, B*), power training (*C, D*), and control (*E, F*) groups. Posttest 2 data for the strength training group involve jump squats at percentages of baseline 1RM. \*Significantly ( $P \leq 0.05$ ) different peak power from baseline. +Approaching significantly ( $P \leq 0.10$ ) different peak power from baseline. xSignificantly ( $P \leq 0.05$ ) different force from baseline. #Significantly ( $P \leq 0.05$ ) different velocity from baseline.



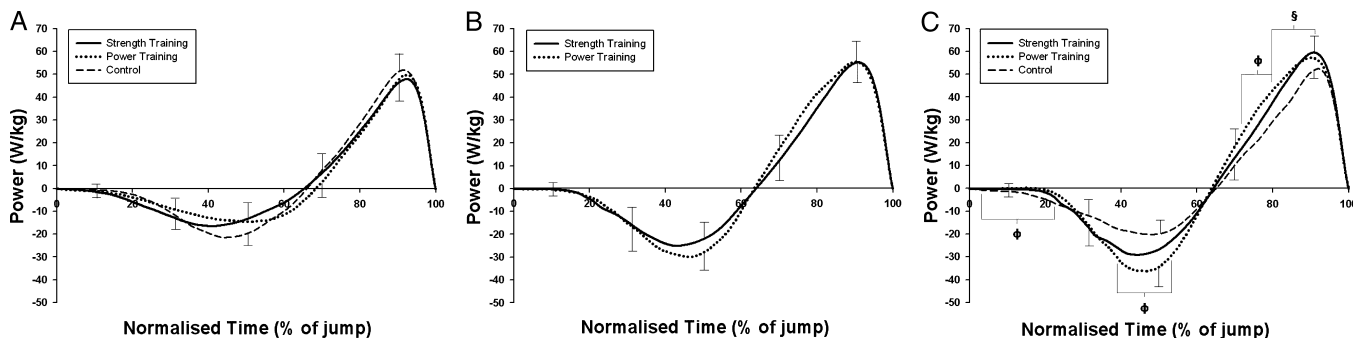


**FIGURE 3**—Between-group differences in the force-velocity (*A–C*) and force-power relationships (*D–F*) for the jump squat throughout the 10 wk of training. Posttest 2 comparisons involve jump squats at percentages of baseline 1RM for the strength training group and percentages of posttest 1RM for the power training and control groups. §Significantly ( $P \leq 0.05$ ) different velocity or power from the control group. ØSignificantly ( $P \leq 0.05$ ) different force from the control group.

countermovement to takeoff). Intraclass test-retest reliability for all EMG variables examined was consistently  $r \geq 0.80$ .

*In vivo* muscle architecture was assessed by B-mode ultrasonography recorded using an ultrasound console (SSD-1000; Aloka, Inc., Tokyo, Japan) with a 7.5-MHz, 9-cm linear probe. The same experienced examiner completed all scans across the baseline, midtest, and posttest occasions. Scans were performed on the VL of the dominant leg, with subjects lying supine and their leg muscles completely relaxed. To assist with acoustic coupling, water-soluble transmission gel was applied to the transducer. Measurements were taken at 50% of the thigh length calculated as half the distance between the centers of the greater tro-

chanter to the lateral condyle of the femur. Longitudinal images were obtained with the transducer oriented parallel to the muscle fascicles and perpendicular to the skin. The impact of the discrepancies in transducer location and orientation on architectural differences observed among the baseline, midtest, and posttest occasions was minimized using several techniques. The location and two-dimensional orientation of the transducer relative to anatomical landmarks were mapped onto a sheet of tracing paper to ensure that the same site was used across all testing occasions. In addition, in an effort to minimize any differences in the three-dimensional orientation of the transducer between testing occasions, the live, on-screen image during midtest and posttest was compared with



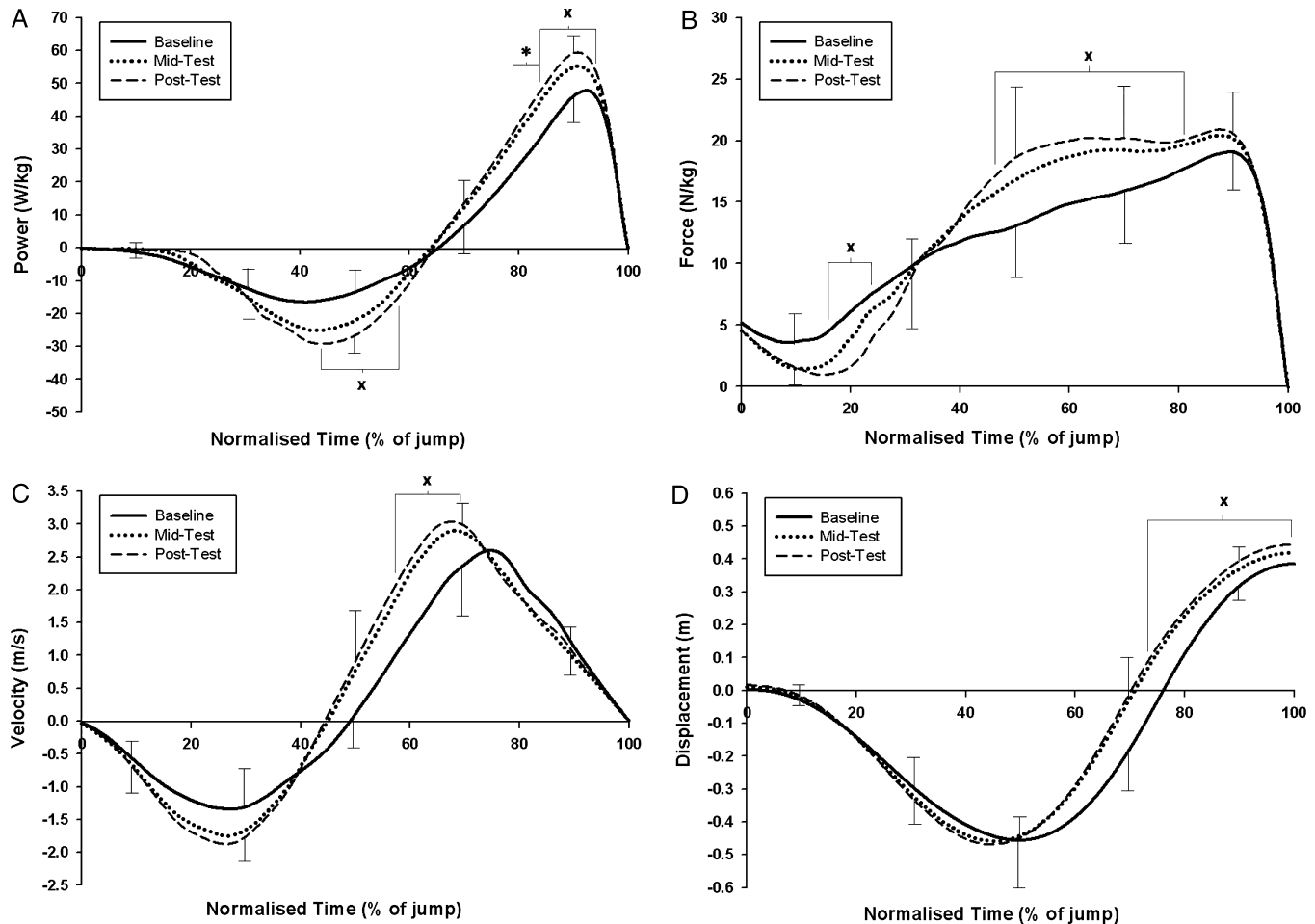
**FIGURE 4**—Comparison of the power-time curve during the 0% 1RM jump squat among the strength training, power training, and control groups at baseline (*A*), midtest (*B*), and posttest (*C*). Normalized time represents the time from initiation of countermovement to takeoff in *A*, *B*, and *C*. ØSignificant ( $P \leq 0.05$ ) difference between power training group and control group. §Significant ( $P \leq 0.05$ ) difference between control group and both strength training and power training groups.

those taken during baseline testing. This allowed the examiner to match the unique features of the ultrasound images (i.e., heterogeneities in the subcutaneous adipose tissue and echoes from interspaces among the fascicles). These procedures have been previously used in research monitoring the impact of training on muscle architecture (3). Images were digitally recorded, and the superficial and deep aponeuroses were identified, with muscle thickness measured as the distance between the aponeuroses (3,21). Pennation angle was defined as the angle at which the fascicles arose from the deep aponeurosis (21). Intraclass test-retest reliabilities for muscle thickness and pennation angle were  $r = 0.95$  and  $r = 0.90$ , respectively. In addition, rater reliability for muscle thickness and pennation angle was  $r = 0.98$  and  $r = 0.93$ , respectively.

Two-dimensional motion analysis was used to evaluate the movement mechanics during the squat 1RM and jump squats. A digital video camera (25 Hz; MV830i; Canon Australia Pty Ltd., North Ryde, NSW, Australia) was positioned 3.1 m from the subject, perpendicular to the subject's sagittal plane. Dartfish software (Version 4.5 ProSuite;

Dartfish, Sydney, NSW, Australia) was used to analyze the movements. The video footage was deinterlaced into fields, yielding a 50-Hz sampling frequency. During the jump squats, the minimum and maximum knee, hip, and ankle joint angles were assessed. These values were recorded at two distinct time points—the transition between eccentric and concentric phases (minimum) and the last field before takeoff where the foot was still in contact with the force plate (maximum). The minimum knee, hip, and ankle angles were also assessed during the squat 1RM representing the relative joint angles at the transition between the eccentric and concentric phases. Intraclass test-retest reliability and rater reliability for all joint angles assessed using motion analysis were consistently  $r \geq 0.92$  and  $r \geq 0.93$ , respectively.

Whole-body composition was assessed using DEXA scans (Discovery A; Hologic, Inc., Bedford, MA). In addition to calculation of percent fat on basis of the ratio of fat mass to total mass of the full body, tissue compositions of the legs were analyzed as individual segments. The leg segment included the foot, shank, and thigh and was delineated



**FIGURE 5**—Training-induced changes to the power-time (A), force-time (B), velocity-time (C), and displacement-time (D) curves for the 0% 1RM jump squat in the strength training group. Normalized time represents the time from initiation of countermovement to takeoff for power and force (A and B) and from initiation of countermovement to peak displacement for velocity and displacement (C and D). \*Significant ( $P \leq 0.05$ ) difference between baseline and both midtest and posttest. xSignificant ( $P \leq 0.05$ ) difference between baseline and posttest.

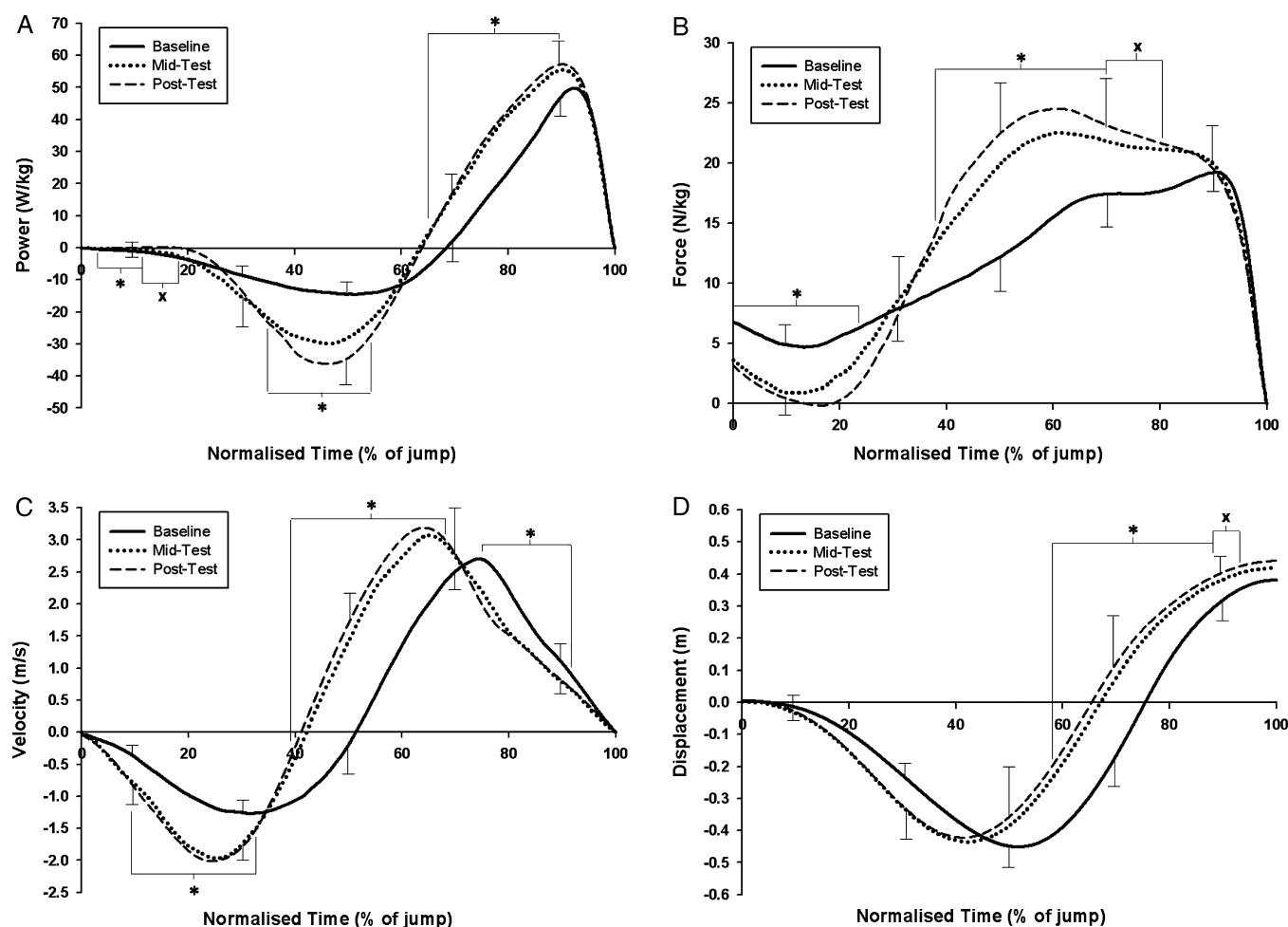
from the rest of the body by a diagonal line running through the neck of the femur, adjacent to pelvis but excluding any mass associated with the pelvic region. Lean mass of the leg was defined as the mass of the lean tissue averaged between the left and right legs. Intraclass test-retest reliabilities for percent body fat and lean mass of the leg were  $r = 0.99$  and  $r = 1.00$ , respectively.

**Statistical analyses.** A general linear model with repeated-measures ANOVA followed by Bonferroni *post hoc* tests was used to examine the impact of training on performance variables and to determine whether differences existed between the groups at baseline, midtest, and posttest. *t*-tests were used for comparisons between variables at baseline and posttest for the control group as well as for comparison of variables between strength training and power training groups at midtest. Statistical significance for all analyses was defined by  $P \leq 0.05$ , and results were summarized as means  $\pm$  SD. Estimated effect sizes (ES) of  $\eta^2 = 0.357$  and  $\eta^2 = 0.300$  at observed power levels of 0.737 and 0.626 for maximum peak power relative to BM

after training existed for strength training and power training groups, respectively. In addition, an estimated ES of  $\eta^2 = 0.420$  at an observed power level of 0.841 existed for the comparison of 1RM/BM at posttest between the strength training and power training groups. Mean ES were also calculated to examine and compare the practical significance of the performance improvements among the experimental groups (6). A statistical software package (SPSS version 13.0; SPSS, Inc., Chicago, IL) was used to perform all statistical analyses.

## RESULTS

**Athletic performance.** No differences in 1RM, 1RM/BM, or maximal isometric force existed among the three groups at baseline. The strength training program resulted in significant improvements in these variables at both midtesting and posttesting occasions (Table 1). After training, the strength training group had significantly greater 1RM and



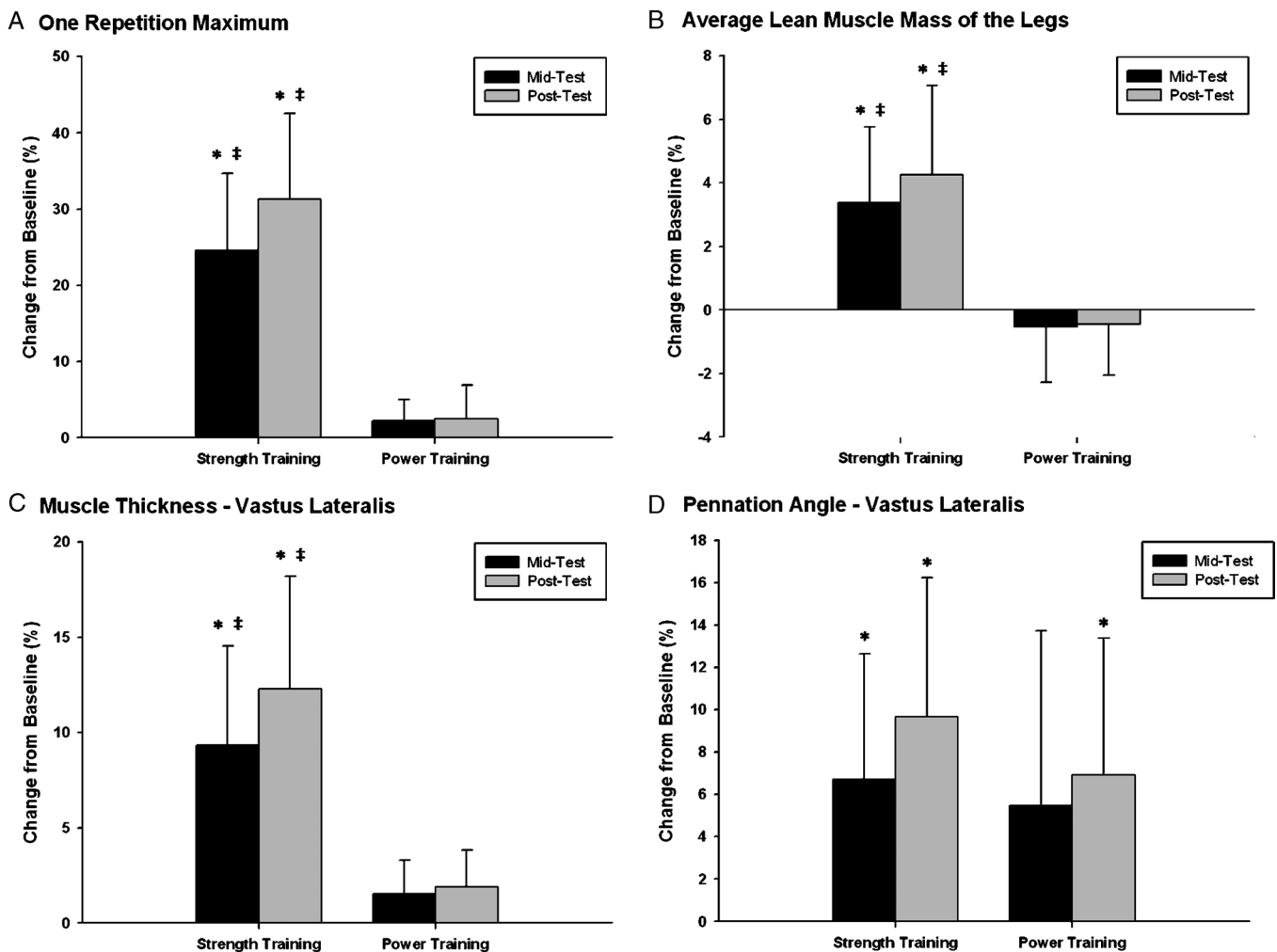
**FIGURE 6**—Training-induced changes to the power-time (A), force-time (B), velocity-time (C), and displacement-time (D) curves for the 0% 1RM jump squat in the power training group. Normalized time represents the time from initiation of countermovement to takeoff for power and force (A and B) and from initiation of countermovement to peak displacement for velocity and displacement (C and D). \*Significant ( $P \leq 0.05$ ) difference between baseline and both midtest and posttest. xSignificant ( $P \leq 0.05$ ) difference between baseline and posttest.

1RM/BM than both the power training and control groups. Significant changes in jump performance were evident after training in both strength training and power training groups (Fig. 1). Training also resulted in significant within-group changes in a multitude of jump performance variables between baseline and midtest or between baseline and posttest sessions for both the strength training and power training groups (Table 2). Before training, no differences in sprint performance were evident among the three groups (Table 3). Training resulted in a significant change in the 40-m time between baseline and posttest for both the strength training and power training groups. No significant differences in sprint performance existed between the strength training and power training groups after training (Table 3).

**Force-velocity relationship.** A multitude of training-induced changes to the force-velocity and force-power relationships during the jump squat was evident after training

in both strength training and power training groups (Fig. 2). Between-group comparisons revealed no significant differences between the strength training and power training groups at baseline as well as after training despite the very different training stimulus (Fig. 3).

**Jump mechanics.** No significant differences existed between the groups in either the minimum or maximum hip, knee, or ankle joint angles during the 1RM and 0% 1RM jump squat at baseline, midtest, or posttest. Training did not result in any changes to the joint angles either at the transition between the eccentric and concentric phases or at takeoff during the 0% 1RM jump squat for either the strength or the power training groups. Furthermore, no within-group differences were evident in the knee, hip, or ankle joint angle at the transition between the eccentric and concentric phases during the 1RM. Investigation of the power-time curve throughout the 0% 1RM jump squat revealed no between-group differences at either baseline or midtesting



**FIGURE 7**—Percent change from baseline at both midtest and posttest in the following: 1RM in the back squat (*A*), average lean muscle mass of the legs assessed through DEXA scans (*B*), muscle thickness of the VL assessed through ultrasound images (*C*), and pennation angle of the VL assessed through ultrasound images (*D*). No significant differences existed between the groups at baseline. \*Significant ( $P \leq 0.05$ ) change from baseline. ‡Significant ( $P \leq 0.05$ ) difference between strength training group and power training group.

TABLE 4. Isometric strength and muscle activation factors assessed throughout the 10 wk training program.

|                                            | Strength Training Group |               |                 | Power Training Group |                |                | Control Group |               |
|--------------------------------------------|-------------------------|---------------|-----------------|----------------------|----------------|----------------|---------------|---------------|
|                                            | Baseline                | Midtest       | Posttest        | Baseline             | Midtest        | Posttest       | Baseline      | Posttest      |
| Isometric squat                            |                         |               |                 |                      |                |                |               |               |
| Peak force (N·kg <sup>-1</sup> )           | 34.0 ± 6.2              | 40.1 ± 3.6*   | 41.3 ± 4.3*,†   | 35.1 ± 7.2           | 35.7 ± 7.1     | 36.0 ± 6.5     | 31.2 ± 4.5    | 30.9 ± 4.4    |
| RFD (N·kg <sup>-1</sup> ·s <sup>-1</sup> ) | 26.5 ± 8.9              | 29.1 ± 10.2   | 35.4 ± 6.0      | 33.5 ± 10.7          | 42.3 ± 7.3     | 49.7 ± 16.0*   | 40.9 ± 17.1   | 41.2 ± 15.4   |
| Time to peak force (% of IS)               | 51.3 ± 10.1             | 45.9 ± 13.2   | 42.6 ± 11.9     | 51.0 ± 14.3          | 45.8 ± 15.8    | 38.2 ± 15.1    | 48.2 ± 22.2   | 46.0 ± 12.4   |
| Muscle activation                          |                         |               |                 |                      |                |                |               |               |
| AvgIEMG IS (MVC)                           |                         |               |                 |                      |                |                |               |               |
| VM (μV)                                    | 57.6 ± 20.2             | 71.6 ± 20.3   | 81.1 ± 14.2*    | 69.0 ± 28.9          | 82.3 ± 44.7    | 81.7 ± 28.2    | 79.6 ± 17.1   | 78.5 ± 18.2   |
| VL (μV)                                    | 59.9 ± 15.0             | 57.0 ± 14.7   | 59.5 ± 9.6      | 73.3 ± 39.0          | 72.6 ± 42.8    | 71.2 ± 28.3    | 71.3 ± 11.9   | 70.0 ± 23.1   |
| BF (μV)                                    | 13.6 ± 6.5              | 15.5 ± 15.3   | 13.6 ± 10.5     | 18.6 ± 14.2          | 14.3 ± 9.4     | 14.8 ± 10.2    | 20.3 ± 11.1   | 20.9 ± 12.0   |
| AvgIEMG 0% 1RM JS                          |                         |               |                 |                      |                |                |               |               |
| VM (% MVC·s <sup>-1</sup> )                | 102.3 ± 27.1            | 111.9 ± 40.3  | 110.7 ± 47.0    | 108.8 ± 29.0         | 121.9 ± 52.3   | 115.3 ± 19.0   | 81.3 ± 13.6   | 84.6 ± 23.2   |
| VL (% MVC·s <sup>-1</sup> )                | 100.9 ± 15.3            | 105.2 ± 25.3  | 105.7 ± 30.4    | 118.8 ± 46.8         | 130.1 ± 59.3   | 106.0 ± 25.7   | 107.1 ± 30.0  | 109.7 ± 28.5  |
| BF (% MVC·s <sup>-1</sup> )                | 105.3 ± 34.5            | 120.3 ± 37.5  | 167.4 ± 99.9    | 100.8 ± 58.8         | 181.7 ± 73.8   | 174.4 ± 90.3   | 103.0 ± 53.2  | 104.9 ± 76.8  |
| Rate of EMG rise 0% 1RM JS                 |                         |               |                 |                      |                |                |               |               |
| VM (% MVC·s <sup>-1</sup> )                | 443.6 ± 231.3           | 629.6 ± 287.6 | 455.5 ± 227.8   | 352.3 ± 174.4        | 725.1 ± 305.9* | 672.5 ± 263.8* | 373.0 ± 209.7 | 386.7 ± 227.4 |
| VL (% MVC·s <sup>-1</sup> )                | 432.8 ± 217.3           | 550.2 ± 202.5 | 558.1 ± 185.3   | 389.6 ± 187.4        | 866.0 ± 672.0  | 654.1 ± 283.0  | 403.6 ± 209.5 | 392.9 ± 146.8 |
| BF (% MVC·s <sup>-1</sup> )                | 378.9 ± 91.9            | 665.7 ± 444.3 | 1058.0 ± 767.7* | 714.5 ± 551.9        | 1180.0 ± 619.0 | 1359.0 ± 984.6 | 449.7 ± 189.1 | 477.2 ± 250.8 |

Average integrated electromyography (AvgIEMG) was assessed during an isometric squat MVC and during the 0% 1RM jump squat. Rate of rise in AvgIEMG also assessed during the 0% 1RM jump squat.

\* Significantly ( $P \leq 0.05$ ) different from baseline.

† Significantly ( $P \leq 0.05$ ) different from the control group.

IS, isometric squat; JS, jump squat.

occasions (Figs. 4A and B). Significant differences were evident at posttest between the control group and both training groups from 80.4% to 90.0% of normalized time (Fig. 4C). In addition, differences between the power training and the control groups were evident from 3.4% to 22.0%, 39.2% to 53.0%, and 72.2% to 80.4% of normalized time (Fig. 4C). Examination of the power-, force-, velocity-, and displacement-time curves throughout the jump revealed significant training-induced changes within both the strength training (Fig. 5) and the power training groups (Fig. 6). For the strength training group, significant differences between baseline and both midtest and posttest existed during the power-time curve from 83.6% to 91.0% normalized time (Fig. 5A). Significant differences between baseline and posttest also existed during the following phases: (a) power: 44.4%–58.0%, 79.6%–83.4%, and 91.2%–93.4% of normalized time; (b) force: 16.0%–23.4% and 47.4%–80.8% of normalized time; (c) velocity: 57.8%–68.4% of normalized time; and (d) displacement: 73.4%–100.0% of normalized time (Fig. 5). For the power training group, significant differences between baseline and both midtest and posttest existed during the following phases: (a) power: 3.6%–11.4%, 35.4%–52.6%, and 64.8%–89.6% of normalized time; (b) force: 0.0%–23.2% and 38.0%–70.0% of normalized time; (c) velocity: 9.4%–32.0%, 39.2%–68.0%, and 75.0%–91.8% of normalized time; (d) displacement: 58.0%–88.2% of normalized time. Significant differences between baseline and posttest also existed during the following phases: (a) power: 11.6%–18.2% of normalized time, (b) force: 70.2%–79.2% of normalized time, and (c) displacement: 88.4%–92.4% of normalized time (Fig. 6).

**Muscle architecture.** No between-group differences existed for muscle thickness, pennation angle, or lean mass of the leg at baseline. After training, the strength training group displayed significant changes in lean muscle mass of

the legs, muscle thickness, and pennation angle (Fig. 7). At both midtesting and posttesting sessions, the change in lean muscle mass of the legs and muscle thickness were significantly greater in the strength training group compared with the power training group (Fig. 7).

**Neural activation.** No between-group differences were evident at baseline in any of the EMG variables assessed (Table 4). After training, the strength training group displayed a significant increase in VM maximal voluntary activation at posttest. No between- or within-group differences were observed in the AvgIEMG during the 0% 1RM jump squat throughout the duration of the investigation. However, the power training group significantly increased the rate of rise in AvgIEMG of the VM during the 0% 1RM jump squat at both midtest and posttest. No within-group changes in any EMG variables were observed for the control group (Table 4).

## DISCUSSION

The primary findings of this investigation were as follows: (a) improvements in athletic performance were similar in relatively weak (squat 1RM/BM =  $1.30 \pm 0.15$ ) individuals exposed to either ballistic power training or heavy strength training for 10 wk (Figs. 1–4 and Tables 2 and 3), and (b) performance improvements were brought about through neuromuscular adaptations specific to the training stimulus (Figs. 2–7 and Table 4).

**Athletic performance.** The current results indicate that short-term exposure to either ballistic power training or heavy strength training involving sports-specific movements elicits improvements in athletic performance in relatively weak individuals. Enhancements in jump squat performance (i.e., jump height, peak and average power,

net impulse, movement velocity, and RFD) were evident after both ballistic power training and heavy strength training (Fig. 1 and Table 2). These findings were similar to previous research involving ballistic power training (10,18,25,39), heavy strength training (34,36,38,39), as well as investigations comparing both training modalities (18,39). The transferability of these training modalities to jump squat performance in relatively weak individuals was very similar, with no significant differences evident between either of the training groups (e.g., change in CMJ height was  $0.06 \pm 0.04$  m for both groups; Table 2). Furthermore, the magnitude of improvements in jump squat performance over a range of loading conditions was very similar between the ballistic power training and heavy strength training groups (Fig. 1). Only the change in peak power output during the 0% 1RM SJ condition differed significantly between the groups after the 10 wk of training (Fig. 1). Thus, it is apparent that both ballistic power training and heavy strength training are equally effective at improving athletic jumping ability in relatively weak individuals. It is important to note that as a result of the 28% increase in squat 1RM after strength training, subjects in the strength training group were also able to produce power over a greater range of absolute loads (i.e., the load–power relationship was extended so that the 80% 1RM load consisted of an additional 25 kg after training). This extension of the absolute load–power relationship has important implications across sports that require athletes to produce power under high loading conditions (e.g., weightlifting, American football, rugby, etc).

A significant change in 40-m sprint performance was also observed after 10 wk of both ballistic power training and heavy strength training modalities in relatively weak individuals (Table 3). The additional improvements in sprint performance observed after ballistic power training (i.e., 20-m, 30-m, and flying 15-m times) may indicate a slightly greater transfer of training to performance. This potential effect is supported by considerably greater ES across the 5-, 10-, 20-, 30-, and 40-m sprints for the power training group; however, there were no significant differences in the magnitude of improvements between the ballistic power training and heavy strength training groups (Table 3). The fact that both ballistic power training and heavy strength training resulted in improved sprint performance is a novel finding of great significance. Previous research has commonly failed to demonstrate any transfer of training to sprint performance after either training modality (18,19,38,39) or observed changes in sprint performance approaching significance after ballistic power training (39). Therefore, the current study is one of the first to observe significant improvement in sprint performance after ballistic power training and especially in response to heavy strength training. The transfer of training to sprint performance in the current study is theorized to be associated with the efficiency of the program design (i.e., highly sports-specific movements, frequency of training,

effective load, repetition, and set and interset recovery parameters) and the relatively nonspecific improvements in muscular function common to previously untrained and relatively weak individuals.

**Mechanisms responsible for improved performance.** Examination of the force–velocity relationship for the jump squat offers some insight into the mechanisms driving athletic performance improvements after ballistic power training and heavy strength training. The ability of such an applied *in vivo* measure of the force–velocity relationship to delineate exact changes to muscle mechanics after training is complicated by a range of factors including mixed fiber composition, muscle architectural characteristics, anatomical joint configuration, levels of neural activation, as well as the complex nature of the jump squat movement (24). Despite these limitations, examination of the force–velocity relationship in sports-specific movements quantifies the ability of the intact neuromuscular system to function under various loading conditions, information essential to understanding muscular function during dynamic athletic movements. The current data indicate that the changes to the jump squat force–velocity relationship were specific to the type of training individuals were involved in (Figs. 2 and 3). Heavy strength training resulted in a significant and quite substantial increase in maximal isometric force, which, in turn, caused a shift in the force–velocity relationship so that subjects were able to generate greater force and power at a specified velocity of movement. This was especially apparent at posttest 2, when the strength training group completed jump squats against percentages of their baseline 1RM (Figs. 2A and B). Ballistic power training also resulted in the ability to generate greater force at a specified movement velocity throughout the force–velocity relationship; however, in contrast to the strength training group, these alterations were not associated with any change in maximal isometric force (Figs. 2C and D). Consequently, there was a tendency for the gradient of the force–velocity curve to decrease in response to heavy strength training and increase after ballistic power training. These changes were similar to those observed by Kaneko et al. (20) in the force–velocity relationship of the elbow flexors after strength training with a variety of different intensities. Despite the different nature of these changes (and the different training stimuli), the jump squat force–velocity and force–power relationships were very similar for both groups after the 10 wk of training (i.e., no significant between-group differences in velocity or power at any of the loads examined).

Other potential mechanisms driving the observed improvements in athletic performance were changes to movement mechanics and stretch-shorten cycle (SSC) function during jumping. Although no significant changes were observed in the minimum and maximum hip, knee, and ankle angles during the 0% 1RM jump squat, significant training-induced changes in displacement, velocity, force, and power were evident throughout the 0% 1RM jump squat for both groups (Figs. 4–6). These changes are theorized to have led

to an optimization of SSC function, which contributed to the enhanced jump performance. Arguably, the primary mechanism driving the enhancement of performance during SSC movements is the attainment of a greater level of force at the beginning of the concentric phase in comparison to the concentric-only movements (5). Both ballistic power training and heavy strength training groups displayed significant increases in force during the eccentric phase and early in the concentric phase of the 0% 1RM jump squat after training (Figs. 5 and 6). This increase was generated by the improved acceleration of the BM during the eccentric phase (i.e., dropped faster into the dip) and, similar to comparisons between SSC and concentric-only movements, resulted in greater net impulse, velocity of movement, power output, and ultimately enhanced jump height after training. Very little previous research exists examining the impact of training on performance variables throughout the movement; however, similar results have been observed after an analogous ballistic power training intervention involving jump squats (9). Comparisons between the ballistic power training and heavy strength training groups revealed no significant between-group differences before or after training in joint angles of the hip, knee, and ankle as well as power output throughout the 0% 1RM jump squat. Despite this, the within-group, training-induced differences in displacement, velocity, force, and power output during the 0% 1RM jump squat were more extensive after ballistic power training (Figs. 4–6). The more prevalent changes observed in this group are hypothesized to be due to slight modifications in jumping mechanics (i.e., a marginally shorter but faster countermovement) and a significant decrease in time to takeoff specific to ballistic power training with jump squats. This hypothesis is supported by a cross-sectional comparison of experienced jumpers (CMJ height > 0.50 m) and nonjumpers (CMJ height ≤ 0.50 m), neither of which were involved in specific jump squat training. Despite displaying superior peak displacement, velocity, force, and power, the jumpers did not show significant differences in any of these variables throughout any other portion of the movement compared with the nonjumpers (9). When this information is coupled with the findings of the current study, it seems that technique modifications specific to training involving jump squats may contribute to improvements in jump performance after short-term ballistic power training.

Changes to muscle architecture were observed at both midtesting and posttesting sessions, highlighting another potential mechanism contributing to the observed improvements in athletic performance. Accompanying the significant increase in squat 1RM, the heavy strength training group displayed a significant change in muscle thickness and average lean muscle mass of the legs after training that was significantly different from the change seen in the ballistic power training group (Fig. 7). These observations were similar to previous assessments of the impact of heavy resistance training on muscle thickness as measured by ultrasound (4,22). Furthermore, Kawakami et al. (22) reported

similar relative changes in muscle thickness (measured by ultrasound) and both muscle volume and CSA (measured by magnetic resonance imaging) after 16 wk of strength training. Thus, although not a direct examination of myofibrillar or whole muscle CSA, the noninvasive measures used in the current study provide a good indication that heavy strength training did elicit a hypertrophic response in relatively weak individuals—a common observation throughout the literature (1,16,22,23). Previous research has established that training-induced increases in CSA are typically accompanied by improvements in maximal force production and maximal muscular power (20,24,33,37,39). The observations of the current study were in support of this body of knowledge and demonstrated that improved athletic performance after heavy strength training in relatively weak individuals was achieved in some part by muscle hypertrophy. In addition, alterations in the pennation angle resulting from both heavy strength training and ballistic power training may have also played a role in the observed performance improvements. Increases in pennation angle have been reported previously after resistance training (1,4,22), although not consistently (3,30), and are believed to have a positive impact on the force-generating capacity of the muscle. As pennation angle increases, more sarcomeres can be arranged in parallel (i.e., more contractile tissue can attach to a given area of an aponeurosis or tendon), and the muscle can therefore produce more force (12,31). In addition, an increased pennation angle allows for muscle fibers to shorten less for a given tendon displacement because of the rotation of pennate muscle fibers during contraction (26). This increases the likelihood that a fiber with a greater pennation angle operates closer to its optimum length and, based on the length–tension relationship, is able to generate more force (26). Therefore, changes to pennation angle may also play a role in improving athletic performance after training.

Perhaps the most significant adaptation driving performance improvements after training in both experimental groups was specific alterations in neural activation. Changes in EMG (indicative of alterations in motor unit recruitment, firing frequency, and/or synchronization) have been extensively reported with improvements in performance after both heavy strength and ballistic power training (13–15,17,25). However, this is one of the first experiments to show training-induced changes in EMG during complex, multijoint, sports-specific movements. In addition, the current study offers vital insight into the differential effect of ballistic power training versus heavy strength training on neural adaptations in such movements. Heavy strength training elicited a significant improvement in maximal muscle activation (i.e., AvgIEMG) during an isometric squat, whereas ballistic power training resulted in significant increases in the rate of EMG rise during the 0% 1RM jump squat (Table 4). Thus, the nature of training-induced changes in neural activation was dependent on the specific stimulus applied by the training. The changes in muscle activation were mirrored by

specific changes in force output during the isometric squat (i.e., heavy strength training resulted in significant improvements in maximal force, whereas ballistic power training significantly increased RFD; Table 4). The current findings are supported by previous research indicating the presence of training-specific changes in neural activation during an isometric knee extension (13,14). Heavy strength training (squatting with 70%–120% 1RM) resulted in changes in maximal integrated EMG that were significantly correlated with changes in maximal force ( $r = 0.66$ ) (13). In contrast, ballistic power training (jump squats with 0%–60% 1RM) resulted in a 38% increase in the rate of EMG rise and a 24% improvement in RFD during isometric knee extension (14). Although the current study cannot delineate whether the observed changes in EMG were brought about through alterations in motor unit recruitment, firing frequency, and/or synchronization, the data indicate that neural adaptations specific to the training stimulus applied contributed to improvements in athletic performance.

## CONCLUSIONS

The transfer of training to athletic performance was very similar for relatively weak individuals exposed to short-term ballistic power training or heavy strength training interventions. Both training modalities resulted in significant improvements in maximal power output, jump height, movement velocity, and sprint performance. However, the mechanisms driving these improvements were specific to the training stimulus these individuals were exposed to. Ballistic power training involving sports-specific movements increased the rate of EMG rise during jumping, which, coupled with slight technique modifications specific to this training stimulus, led to an optimization of SSC function. The power-trained subjects were able to produce more force and increase the RFD, which resulted in greater acceleration and movement velocity in shorter periods. Consequently, athletic performance was improved through enhanced magnitude and RFD, resulting in increased velocity and power being realized. In contrast, performance improvements after heavy strength

training involving sports-specific movements were a result of increased maximal neural activation and muscle thickness. These adaptations increased the contractile capacity of the lower limb musculature, which meant that the BM represented a lower relative load after training. Consequently, strength-trained subjects were able to produce more force and increase the RFD, which resulted in the ability to accelerate their mass to a greater degree in the same period. Therefore, athletic performance was improved through enhanced magnitude and RFD, resulting in increased velocity and power being realized. Not only do these findings provide a deeper understanding of the mechanistic factors driving improvements in performance after various applied sports-specific resistance training modalities, but they have important implications for the design of effective resistance training programs. The ability of heavy strength training to render similar short-term improvements in athletic performance as ballistic power training, coupled with the potential long-term benefits of improved maximal strength, makes heavy strength training a more effective training modality for relatively weak individuals (i.e., previously untrained individuals and novice, youth, or endurance athletes). Thus, an individual does not necessarily need to place a focus on ballistic power training until a solid foundation of strength is obtained (i.e., squat 1RM/BM ratio  $\geq 1.60$ ). Furthermore, these findings have significant implications for our understanding of the influence of strength on power production and athletic performance. When relatively weak individuals improved their maximal strength, the ability to generate maximal power and velocity during athletic movements also increased in the absence of any specific ballistic power training. Therefore, maximal strength (and its underlying neuromuscular characteristics) plays an important role in the production of maximal power and superior athletic performance.

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